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Green Design: Introducing a New Methodology to Increase Environmental Sustainability in Capital Investments at AstraZeneca

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Abstract. The concept of environmentalism is constantly growing as new customer demands and regulations are introduced. Preventive measures on production system design are lacking as the environment is being overlooked in capital investment projects of production equipment and systems. The objective of this study was to explore the need for environmental sustainability in capital investment projects and develop a methodology to implement resource efficiency and circularity. It is identified that the design phase of capital investment projects has a great opportunity for decreasing environmental impact in future operations. However, strategy deployment of environmental sustainability to project level is lacking, justifying the urgency for additional environmental approaches. The Green Design methodology developed introduces environmental sustainability to capital investment projects by utilizing several green-lean and circular tools. The tools are applied through identifying, evaluating, implementing, and follow-up on environmental improvements.

Keywords: Green Design · Capital Investment · Production Equipment · Production System Design

1 Introduction

As the sustainability concept is increasing worldwide, the new environmental paradigm to reduce negative environmental impact implies targeting climate neutral products to be sold on the world markets. However, as the environmental impact share from products gradually decreases, the share from production consequently increases. The production share motivates more focus to be put on designing environmentally sustainable production systems from the start, as found crucial by e.g., [1]. Industrial production continues to increase. A 50% decrease in production resource use is expected by 2050, whereas industrial output is expected to double by the same time [2, 3]. This half resource-double output projection suggests need for increasing production capacity while also motivating industries in becoming more environmentally sustainable regarding production (i.e., not only products). One essential part of environmentally sustainable production systems is

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the invested hardware assets. These are to be designed and operated with a life cycle perspective, hence in a resource efficient and circular way. Approximately 80% of a product's life cycle cost, connecting also to most of the environmental impact, is determined in the early design stages [4]. This statement suggests that only 20% of environmental impact is possible to decrease during the use phase (operations). Not taking assets into account when designing a production system increases the risk of embodied linearity. Instead, circularity is now required to manage cutting resource use while simultaneously increasing output.

The objective of this study was to explore the need for environmental sustainability in capital investment projects and develop a methodology to implement resource efficiency and circularity. This was fulfilled by firstly exploring the current state of environmental sustainability inclusion on capital investment project level and its relation to corporate sustainability strategies at a pharma company. Secondly, by acquiring knowledge on what approaches could be used to increase environmental sustainability through the design phase of capital investment projects.

This study was limited to capital investment projects connected to production system design, i.e., equipment, systems, construction, media, and utilities. Other projects related to supply chain, IT, etc. were excluded.

2 Related Research

This chapter presents the subjects studied through a pre-study and literature review. Focus is on environmental sustainability inclusion in corporations and on operations level, the interconnection between production system design and capital investments, and environmental approaches suitable for production.

2.1 Corporate and Operational Sustainability

Organizations of today are being both externally pressured and internally motivated into taking a more proactive role to adjust towards sustainability in all areas of their businesses. Emerging regulations and consumer environmentalism are identified as two major factors for change [5]. New business models and strategy adjustment towards environmental processes are also required [5]. Production industries have great opportunities in reducing their environmental impact due to being resource and capital intense [6]. As part of operations, large waste generation is a main driver for change [7]. Also, the rising cost on energy and material is becoming a large motivator together with the dependency on non-renewal resources where alternatives are required [6, 7]. If environmental goals are not implemented in corporate strategies, companies are not willing to introduce environmental design activities and dedicate time on those activities [2]. Multiple large Swedish companies are implementing sustainability goals on corporate level, for example by adhering to ISO14001, initiatives on CO_2 reduction by an x percentage by year y, and so on [8]. The goals set are primarily focused on production performance.

Green Performance Map (GPM) was developed based on lean methods to improve environmental performance in production. GPM focus on identifying and improving ingoing and outgoing parameters to a production system. Parameters studied are for

example energy, productive material, emissions, and waste [9]. The purpose is to enable communication and less complex environmental work, and support environmental continuous improvement (*Green Kaizen*) [9, 10]. The method (called Green Kaizen) is today a global standard at several manufacturing and production companies, including AstraZeneca.

2.2 Production System Design and Capital Investments

This paper specifies production system as an interacting combination of people, material, machines, tools etc. designed to work together for a common purpose [11].

Production system design is generally down-prioritized due to it being considered time- and resource consuming [12]. Considering the 80% of embodied environmental issues (commonly embodied carbon) connected to the product design phase [4], it can be anticipated that production system design is essential to reduce emissions and resource consumption during operations. Changes to an up-and-running production system suggests limitations due to the hindsight costs and time restrictions [12]. Production system design and alteration is a task that belongs to the Production Engineering Department executing capital investment projects on equipment, systems, construction, media, and utilities. These projects typically follow a company specific guide based on the production system life cycle [13, 14], see Fig. 1.



Fig. 1. The production system development life cycle

Production system efficiency is necessary to realize customer needs and to stay competitive in a changing market [13]. Not only is operational efficiency necessary, but also the environmental efficiency is critical to stay competitive where customer awareness and requirements increase towards sustainability throughout the product life cycle [15].

Environmentally sustainable production system design implies focusing on the entire life cycle of products including supply chains, equipment, maintenance etc. A holistic life cycle perspective on production system design allows for identifying weak links where focus should be on decreasing the highest environmental impact. The objective when designing a production system to be environmentally sustainable is to increase the environmental performance (e.g., reduced amount of material, increased length-of-life), and minimizing resource and energy consumption [16, 17]. However, a lack in awareness and knowledge of implementation of environmentally sustainable design is identified [2]. Companies generally have difficulties in modifying design activities into *environmental* design activities due to the extra amount of time needed [2]. Despite a significant number of environmental design approaches available, companies lack practical implementation and use [2].

The steps of the production system development life cycle, as illustrated in Fig. 1, follows a generalized project model on capital investment. Fig. 1 suggests that production system design is achieved through capital investments in industry. Capital being the

references to installed/constructed equipment, machines, tools, infrastructure, and facilities. It is suggested that more equipment is being procured and installed in industry due to increased automation and digitalization [18]. Environmental regulations together with resource depletion demands environmental production and waste elimination to enable circular investments [19]. Industrial production organizations, being a giant resource consumer, are required to increase knowledge in environmental design [20]. Increased knowledge suggests increased sustainability and circularity in capital investment is required to manage the environmental paradigm shift together with the digital paradigm shift [18, 20]. Hence, environmental sustainability must be included into traditional capital investment projects and aligned with incentives of profitability and operational efficiency.

Scope 3 emission reporting and EU's Taxonomy are other incentives for transforming into environmentally sustainable capital investments. Scope 3 emissions are emissions as result from assets not controlled or owned by the producing and reporting company. Thereby indirectly affecting organizations value chain impact [21]. The EU Taxonomy is a classification system to bring incentive on organizations' environmental capital expenditure to increase the degree of environmental sustainability in investments. The increase is made by stimulating companies to measure environmental costs of their business and profits derived from environmental capital expenditure [22].

2.3 Green-Lean and Circular Economy

The literature review conducted disclosed two major subjects associated with inclusion of environmental sustainability in production context: green-lean and circular economy. Within these two, several approaches were identified as appropriate for utilizing in the design phase of capital investment projects.

Lean manufacturing, as considered a large influence in the production paradigm, focuses on increasing productivity through waste elimination [15]. Companies need not only focus on improving productivity, but also environmentally. The integration of environmental sustainability, or *'green'*, and lean is increasing to include full potential of waste handling [15, 23]. Several green-lean approaches exist to increase sustainability together with productivity in industry. The define-measure-analyze-improve-control (DMAIC) method is suggested to improve production processes by eliminating defects, where environmental impact is considered a defect [24, 25]. Several approaches were proposed for identifying, evaluating, and assessing environmental impact for decision making and optimizing processes. Among them, three methods were identified as appropriate for this study. Life cycle assessment (LCA) to quantify environmental impacts of a process/equipment [25, 26]. Eco-efficiency indicators to classify areas of environmental impact [26, 27]. Environmental value stream mapping (E-VSM) to map eco-efficiency indicators for increased understanding of major environmental impacts and thereby prioritization [28, 29]. However, it is identified in research that extensive time consumption is seen as a major drawback when utilizing these approaches. This drawback must be compensated by realizing the opportunities with environmental impact and waste handling [25].

Preventing waste in total is identified by Ellen MacArthur Foundation [30] as the next sustainable paradigm shift. This shift will challenge the conventional 'take-make-waste'

handling of resources by recirculating resources back into the production system through *circular economy* [30]. The concept of circular economy builds upon three principles, (1) eliminate waste and pollution, (2) circulate products and materials, and (3) regenerate nature [30]. The literature review disclosed approaches for supporting the three circular principles through efficient use of resources and materials, and waste reduction. Four approaches were identified as appropriate for this study. Waste analysis for analyzing types of wastes included in a process/project and their opportunities for avoidance and recirculating [31]. End-of-life decision making by determining type of recirculation based on functionality of material/equipment [32]. Product-service systems introduces a leasing-based agreement on equipment where the supplier has a clear takeback initiative [33]. The reduce-reuse-recycle (3R) principle suggests a hierarchical model for design decisions where reducing material use is priority [34]. Introducing resource optimization and waste reduction aligned with circular economy is highlighted to stay competitive in production industry [34]. It can also be argued required to achieve the half resource-double output projection in industry [2, 3].

3 Methodology

This study was carried out primarily as a case study at AstraZeneca Södertälje, Sweden. A first version of a methodology on implementing environmental sustainability in capital investment projects had been developed the previous year at AstraZeneca.

To establish the foundation for the industrial case study, a pre-study was conducted involving search for key literature connected to relevant topics (see Sect. 2, *Related Research*). After conducting the pre-study and setting the scope for the study, an interactive case study was conducted at the case company. In addition, interviews were made with three selected large Swedish producing companies to get more insights of the problem area on the extent sustainability is included into corporations. The study was designed as an interactive case study consisting of multiple units of analysis. The purpose was to investigate both academic research and multiple companies' environmental work to produce approaches that could be implemented at the case company. Empirical data was collected through several semi-structured interviews, analysis of internal company documents, observations, informal discussions, and participation in workshops at the case company where the new methodology was tested. The empirical data collection record of the case study is presented in Table 1. For confidentiality reasons, no other companies than AstraZeneca are disclosed.

In addition, a structured literature review was conducted based on the pre-study to identify green-lean and circular economy approaches suitable for production system design. The literature review was conducted extracting academic references from external databases using search strings. Keywords connected to green-lean and circular economy as identified during the pre-study were used in the search strings, e.g., greenhouse gas emissions, production, cradle-to-cradle, and equipment. A total of 57 references were identified and analyzed as part of this study with a majority within the 2010-current time range.

Inductive thematic analysis was used in the study because of its relevance when analyzing large amount of qualitative data [35], and because of the nature of interviews

Table 1. Empirical data collection record

Code	Type of industry	Type of study	No. of interviews	Interviewee position
AstraZeneca	Pharmaceutical	Interview, workshop, document analysis	12	Front-end study manager, project manager, process engineer, associate director of procurement, engineer manager, environmental compliance lead, sustainability lead, senior lean coach, industrial PhD
University	Academia	Workshop	3	Researchers
Company A	Automotive	Interview	2	Expert environmental sustainability, sustainability engineer
Company B	Automotive	Interview	1	Property- and environmental manager
Company C	Machine components	Interview workshop	1	Knowledge development manager

being semi-structured. The analysis was divided accordingly to the research methodologies used. The pre-study was conducted through gathering of key articles and a backwards snowballing method. References from the pre-study and literature review were thematically coded into themes. These concerned corporate sustainability, production system design, green design, capital investments and acquisitions, requirements and specifications, green-lean approaches, and circular economy. The data collected through the case study was coded into themes concerning environmental goals, circularity, green tools, goals on investment, critical factors for investment, and green requirements.

The data found through empirical research and literature review was used to construct the conceptual methodology for inclusion of environmental sustainability in capital investment projects. The development of the methodology was realized through a plan-do-check-act (PDCA) cycle where concepts were elaborated and verified by employees with typical roles in capital investment projects through workshops.

4 Empirical Results

4.1 Corporate Sustainability Inclusion in Studied Companies

The case company, as well as the complementing companies selected for the study, had implemented sustainability strategies on corporate level to reduce emissions, waste, energy etc. up to a future point in time. However, only one company demonstrated having a goal or KPI for capital investment projects as part of their environmental sustainability strategy. None of the four companies disclosed environmental sustainability to be a critical factor when deciding on capital investments. See Table 2 for a summary of their environmental sustainability inclusion.

Table 2. Environmental sustainability inclusion at companies interviewed

Code	AstraZeneca	Company A	Company B	Company C
Environmental goals	98% reduction in CO_2 emissions at own sites and fleet by 2026. 90% reduction in CO_2 in entire value chain by 2045 with 10% carbon capture.	50% reduced CO_2 emissions between 2015–2025 through electrified vehicles, renewable fuels, and waste elimination.	Goal on waste reduction through increased reuse and recycling initiatives.	Net zero CO_2 emissions in all sites by 2030. Net zero supply chain by 2050.
Steps taken towards goal	Green Kaizen, *Green Design Event*, KPIs tracking emissions and resource usage.	GHG protocol for calculating climate impact. Lower energy consumption on new equipment.	Measuring material use efficiency and CO_2 emissions. IMDS for new products.	Introducing hydropower, solar power, and geothermal energy. Reusing honing oil.
Goals or KPIs on capital investment projects	95% of capital projects involving a design phase should perform a *Green Design Event*.	*Not specified*.*	Not implemented.	*Not specified*.*
Critical factors for investments	Cost, usability, safety.	Cost.	Cost and quality.	*Not specified*.*

(*continued*)

Table 2. (*continued*)

Code	AstraZeneca	Company A	Company B	Company C
Environmental requirements	Requirements connected to by-products. Larger suppliers must be certified (ISO14001).	Requirements connected to CO_2 emission and energy consumption. Larger suppliers must be certified (ISO14001).	Suppliers must be certified (ISO14001). No other requirements concerning sustainability or environment.	Strategy including future environmental requirements to achieve environmental goals, such as certification (ISO14001 and ISO50001).
Circularity	No current processes for reuse or refurbishment. Recycling is done, certain equipment is sold.	Certain wastes and water are being recycled or reused. No clear strategy for inclusion of CE.	New business model including circularity in form of offering renovation kits for products. Also investing in repairable systems in operations.	Laser treatment for repairing products. Refurbishment is offered on certain products.
Environmental tools	Green kaizen, *Green design Event*.	Environmental checklist, Green kaizen.	GPM.	Simplified LCA.
SHE handling	Larger focus on safety and health.	Larger focus on safety and health due to regulatory reasons.	Larger focus on safety and health due to regulatory reasons.	*Not specified**.

*No knowledge was found on the matter.

4.2 Green Design – A Conceptual Methodology for Capital Investments

The structured literature review concluded several approaches as appropriate to increase environmental sustainability in production companies. Four green-lean approaches were identified as appropriate for usage in production system design, and thereby within capital investments. (1) green lean six sigma and the DMAIC method, (2) LCA, (3) E-VSM, and (4) eco-efficiency indicators and assessment. Also, four circular approaches were identified, (1) waste analysis, (2) end-of-life path, (3) product-service systems, and (4) the 3R principle. All approaches are frequently occurring in theory and are concluded to be appropriate for use in producing companies (see Sect. 2 *Related Research*).

A conceptual methodology called *Green Design* was developed and launched as a method at AstraZeneca, Sweden Operations in Södertälje, Sweden. The intention of

Green Design is to decrease the gap in project sustainability and increase possibilities for environmental sustainability inclusion throughout the organization, thereby also increasing the sustainability culture.

The Green Design methodology was developed based on the Green Performance Map (as a part of the method Green Kaizen) [9, 10] and DMAIC [24, 25]. The purpose was to produce tailored environmental sustainability improvements for capital investment projects in the production context (factory). The name *Green Design* was aligned with the name of the environmental improvement method called *Green Kaizen*, although targeting the design phase instead of the operations phase. The conceptual methodology, referred to as the Green Design methodology, was developed to support project teams in identifying, evaluating, implementing, and follow-up on environmental improvements in production investment projects. It was driven by the goal to make the investments as environmentally sustainable as possible throughout the life cycle of the investment (equipment, machine etc.).

Prior to this study, a first stage of the Green Design methodology had been introduced to new capital investment projects through a workshop called Green Design Event. During the workshop, the project team answered a set of questions to identify environmental improvements for the project. However, in this early pilot stage, projects were left without a clear process on how to proceed towards implementing the identified and prioritized improvements.

The case study followed the update of the Green Design Event, changing to fewer and more concise questions. The questions formulated were derived from the four green-lean and four circular approaches as identified in the structured literature review conducted. The questions were further divided into four categories for input into the production system, and four categories for output as derived by GPM [9]. One category was changed to better fit AstraZeneca's need. The four input categories are (1) energy, (2) value-adding material, (3) equipment and facilities (originally non-value adding material), and (4). The four output categories are (1) emissions (air/noise), (2) productive output, (3) non-productive output, and (4) emissions (water/soil). After each Green Design Event (executed as a 2-h workshop with a broad participation within different knowledge areas), identified suggestions were to be evaluated. Evaluation was based on environmental impact and project impact, i.e., how specific suggestions were impacting the project scope, budget, timeframe, and resources. As a major complement (and part of the conceptual methodology), tools for evaluation were developed primarily based on a simplified LCA, E-VSM, and eco-indicators. These as suggested by the literature review. Post evaluation, the improvements were reviewed by the project sponsor and steering committee for further decision on implementation into the project scope. Also, the improvements were included in the requirement specification for the equipment suppliers where necessary. The Green Design methodology as developed in 2022 is illustrated in Fig. 2.

Green Design was inspired by the Green Kaizen method and follows a stepwise methodology derived from DMAIC as suggested in the literature review; (1) defining the scope of the project and identify environmental problems and improvements, (2) measure and evaluate suggestions for overcoming the problems, (3) implement improvements, and (4) ensure improvements having good effect on business, and follow-up. The first

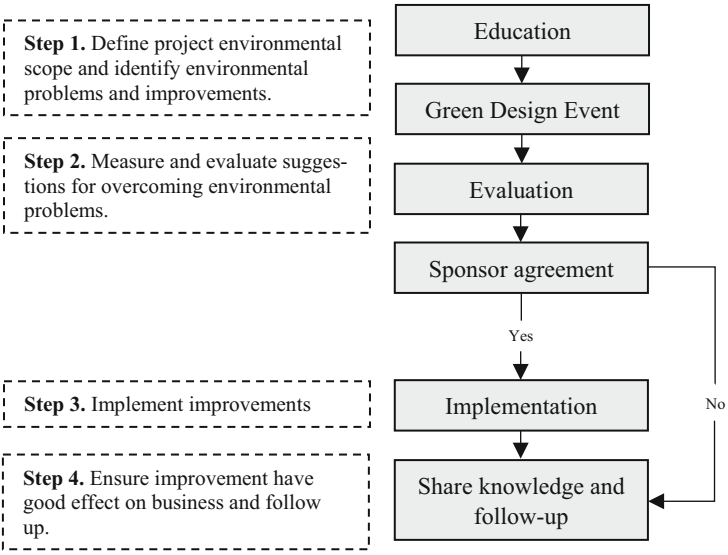


Fig. 2. The Green Design methodology developed at AstraZeneca

step (1) in the methodology was expanded to also include education due to identified lack of knowledge on sustainability. Lack of knowledge was seen as hindering the project teams' possibility to identify environmental improvements. The Green Design Event itself was updated using less, and more generalized questions to enable further utilization areas than prior event. Questions were connected to waste analysis, end-of-life path decision making, product-service system, and the 3R principle as suggested by the literature review. Also, questions concerning maintenance and construction was included to fit AstraZeneca's need. Step two (2) was implemented by including existing, and developing new, tools. These gave the project team possibility to calculate the project impact, i.e., scope, budget, time, and resources. Also, estimated environmental impact was calculated by measuring CO_2, energy usage, water usage and waste generation. Step three (3) on implementation dealt with identified and evaluated environmental improvements together with project impact. Some which could require anchoring with sponsor and possibly steering committee to ensure agreement on implementation. The fourth (4) and last step intended to ensure good impact on business through follow-up as well as an opportunity to enhance the Green Design methodology through sharing knowledge and experiences.

The updated Green Design Event was tested on three ongoing projects at AstraZeneca Södertälje. The first test was on a project concerning cooling equipment where ten environmental improvements were identified and taken for further evaluation. The second test was on a project concerning capacity increase of a cleaning process where eleven environmental improvements were identified and taken for further evaluation. The third and last test was on a project concerning decontamination equipment where 12 environmental improvements were identified and taken for further evaluation. Due to the length of the capital investment projects, assessing the results thoroughly has not been possible as

part of this study. However, workshops and discussions were held with project managers and teams on the continuous work required for implementation. This brief assessment concluded that the tools provided in the Green Design methodology were practical in usage since they provide project teams with possibilities for simplified LCA's amongst others. The results functions as a basis for scope changes, improving the potential of achieving more environmentally sustainable investments in the end.

5 Analysis

5.1 Corporate Sustainability Inclusion

All companies interviewed in this study have implemented sustainability goals and taken steps towards environmental initiatives motivated by the ongoing discussions on sustainability in industry worldwide. However, AstraZeneca was the only company of the four studied that have implemented a goal on capital investment project level. Engineering management decided on a KPI stating that 95% of all capital investment projects should identify environmental improvements. Actual implementation of improvements was not, however, ensured by the KPI. Identification of improvements was rather ensured. Identifying environmental necessity on corporate level implies motivation to include sustainability strategies at this level. However, the gap in knowledge as identified by Rossi et al. [2] demonstrates together with the AstraZeneca KPI that corporate strategies were not fully deployed throughout the organization. Capital investment project organizations were overlooked. All companies studied had implemented environmental work on operational level of the organization, implying that sustainability strategies were deployed to operational level. The 80% reduction opportunity when improving environmental sustainability in capital investment project design [4] justifies sustainability strategy deployment in three stages: corporate sustainability, capital investment project sustainability, and operational sustainability, as illustrated in Fig. 3.

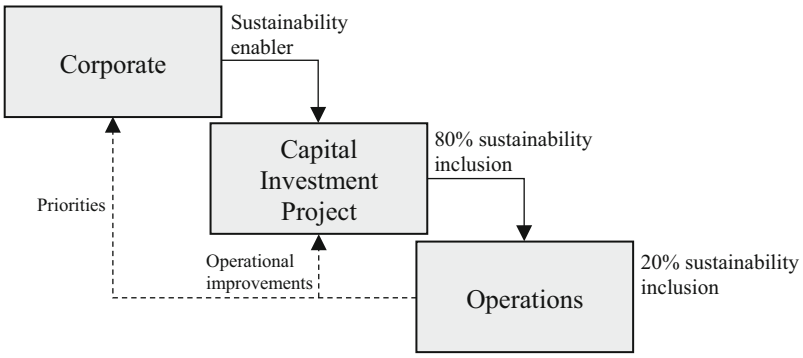


Fig. 3. Sustainability strategy deployment

Targets are set for implementing environmental sustainability at stage 1 – corporate level. This stage sets the strategy towards change, 100% of sustainability inclusion is

enabled in this stage. The next step is to derive the 100% into actual work. Following the concept of 80/20 environmental impact connected to design/operations. 80% of opportunity to reduce impact is in stage 2 where production systems are being designed through capital investment projects. Stage 2 decides the amount of embodied environmental issues. Stage 3 is the remaining 20% where adjustments can be made to up-and-running equipment, systems etc. to reduce energy consumption, emissions etc. (optimization). It is important to realize the operational and capital adjustments needed towards sustainability when deciding at corporate level, where priorities and improvements are fed by operations.

Initiatives to improve sustainability should be cascaded throughout the organization to assure production system design in form of capital investment is reached. It could be assumed that most of the capital investment projects are brownfield, i.e., alterations made to up-and-running production systems, based on strategic re-investments made in companies. Changes must be made in the design phase of all projects, not only greenfield (new, with no or limited restrictions), to assure that embodied environmental impact is decreased.

5.2 Green Design

AstraZeneca has since this study performed additional Green Design events and the Green Design methodology has begun spreading as a best practice to include other types of projects (IT, supply chain etc.). AstraZeneca has appointed an operative team of four engineers called the Green Design team with the task of developing and managing the methodology further. The team is also engaged in a research project with KTH Royal Institute of Technology together with two large manufacturing corporations and a supplier of production equipment. The analysis made in this paper draws conclusions from both work made in the study as well as work made post the study.

The possibility to introduce a methodology to identify, evaluate, and implement is supported by both academic best practices together with empirical results on industrial level. The new Green Design methodology allows AstraZeneca to take the next step towards inclusion of sustainability also on project level where identified improvements require support in assuring implementation. Simplified approaches for support, such as simplified versions of LCA, VSM etc., are preferable due to the extensive work required in capital investment projects. Also, utilizing already applied tools within the organization was preferable to achieve a higher degree of acceptance within multiple departments. Tests conducted on developed tools suggested not only decrease in energy, waste, water, and/or CO_2 consumption, but also decreased operational costs as a result. The outcome of the initial events implies that expertise within the concerned project area is necessary when identifying environmental improvements. In several Green Design events, most valuable input came from experienced process operators, process engineers, and service engineers, who are the people working closest to the processes. However, a facilitator is still needed to steer discussions towards focusing on sustainability and environmental improvements. The facilitator should be generally skilled in operational processes to transpose technical solutions into fitting environmental improvements. The Green Design methodology in general was deemed an appropriate way of working with sustainability in capital investment projects regardless product or industry. In addition,

Green Design has contributed to an increased environmental corporate culture where sustainability is a larger part of capital investment projects.

6 Conclusion

This study identifies four green-lean, and four circular approaches as appropriate for usage in capital investment projects. The eight approaches were used when developing a methodology for including environmental sustainability in capital investment projects at AstraZeneca. The methodology was developed by utilizing the DMAIC method where the methodology follows the four framework steps (1) identification, (2) evaluation, (3) implementation, and (4) follow-up. The first two steps utilize approaches identified in literature to produce environmental improvements and evaluate these using simplified tools which supports project managers where time is often restricted. The latter two steps have been tested post this study but with limited number of results. No conclusions have been made to these latter two steps at this point.

To summarize, environmental sustainability is implemented on corporate strategy level and on operational level in the companies studied. However, in relation to capital investment projects, environmental sustainability is lagging towards other corporate focus areas. The vision of becoming a green organization is recognized on high managerial level, but not perceived as clear on project level. The recognition on project level is required based on data collected in academic references and empirically to enable future resource efficient and circular operations. This study recognizes multiple approaches for improving environmental sustainability on project level, suggesting companies to focus on the design phase of capital investment projects as a way. Looking at capital investment projects, cost is still the most important factor when deciding upon investments. For future research and due to the importance of cost at most companies, it would be most beneficial with more knowledge and tools that supports the decision-making concerning the tradeoff between cost and environmental aspects. Such tools should also include the operational costs to make it possible to compare the case of making environmental improvements vs making investments. It could be assumed that the new EU Taxonomy will support an increased focus on making more sustainable investments.

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